Calc 1 Lecture 25: The Fundamental theorem of Calculus

<u>Derivatives</u> and <u>integrals</u> are the fundamental operations of calculus.

It turns out they are very closely related to each other.

This may seem surprising at first — what do slopes (derivatives) and areas (integrals) have to do with each other?

But we can instead use the framework of <u>rates of change</u> and this connection becomes clearer.

Remember: if $f(t)$ represents velocity as a function of time (assume for now positive)

$$\int_a^b f(t)\,dt = \text{displacement (overall change in position)}$$

$$\int_a^b f(t)\,dt = \lim_{n\to\infty}\left(\sum_{i=1}^{n} \underbrace{f(t_i)}_{\text{velocity}} \cdot \underbrace{\Delta t}_{\text{time}}\right) \quad \text{velocity} \cdot \text{time} = \text{distance}$$

This is true more generally...

if $f(t)$ = rate of change of $F(t)$ $\left(\begin{array}{l} F' = f \Rightarrow \\ F \text{ is an antideriv. of } f \end{array}\right)$

$$\int_a^b f(t)\,dt = \left.F(b) - F(a)\right] \text{ overall change}$$

$$\int_a^b f(t)\,dt = \lim_{n\to\infty}\left(\sum_{i=1}^{n} \underbrace{\underbrace{f(t_i)}_{\approx\frac{\Delta F}{\Delta t}}\,\Delta t}_{\approx\,\Delta F}\right) \quad \text{adding up small changes in } F = \text{net change}$$

ΔF = "small change"

Fundamental Theorem of Calculus part 2:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where $F(x)$ is an antiderivative of f

Example $\int_0^1 x^2 dx$ the easy way.

Step 1: find an antiderivative of $f(x) = x^2$

$$F(x) = \frac{x^3}{3}$$

Step 2: use FTC 2

$$\int_0^1 x^2 dx = F(1) - F(0) = \frac{1^3}{3} - \frac{0^3}{3} = \boxed{\frac{1}{3}}$$

Example Evaluate $\int_0^2 (3e^x - x^3)\,dx$

antiderivative of $\underbrace{3e^x}_{e^x} - \underbrace{x^3}_{\frac{x^4}{4}}$? $\quad F(x) = 3e^x - \frac{x^4}{4}$

$$= F(2) - F(0) = \left(3e^2 - \frac{2^4}{4}\right) - \left(3\underbrace{e^0}_{1} - \frac{0^4}{4}\right)$$

$$= 3e^2 - 4 - 3 = \boxed{3e^2 - 7}$$

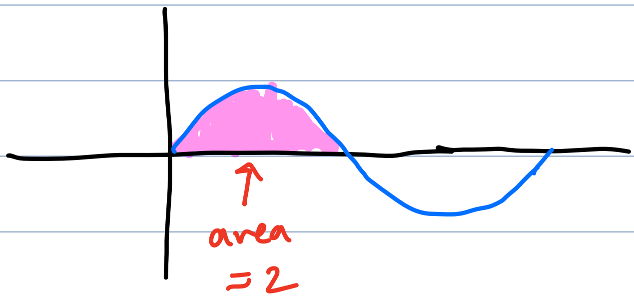
Notation: $F(x)\Big|_a^b = F(b) - F(a)$

e.g. $x^4\Big|_0^2 = 2^4 - 0^4$

Example Find $\displaystyle\int_0^{\pi} \sin(x)\,dx$

Answer: $= -\cos(x)\Big|_0^{\pi} = -\cos(\pi) - (-\cos(0))$

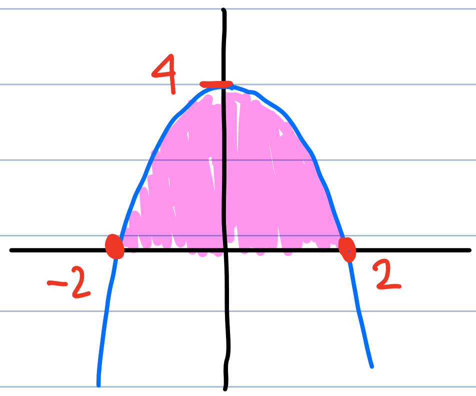
$-(-1) - (-1) = \boxed{2}$



Example Find the area bounded between $y = 4 - x^2$ and the x axis.

x-intercepts: $4 - x^2 = 0$

$x^2 = 4 \Rightarrow x = \pm 2$



area $= \displaystyle\int_{-2}^{2} (4 - x^2)\,dx$

by symmetry

$= 2\displaystyle\int_0^2 (4 - x^2)\,dx = 2\left(4x - \frac{x^3}{3}\right)\Big|_0^2$

total area = 2 · (area on right)

$= 2\cdot\left(8 - 8/3\right) = \boxed{\frac{32}{3}}$

Fundamental Theorem of Calculus part 1

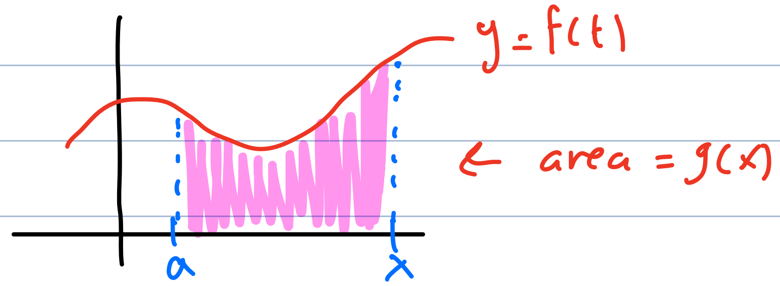
Let $f(t)$ be some continuous function. Let a be some fixed number

Consider $g(x) = \int_a^x f(t)\,dt$

a fixed lower limit
x variable upper limit

two interpretations.

- Area: $g(x)$ is the area under the curve $y=f(t)$ from a to x



- Rates of change:

$g(x) = F(x) - F(a)$, where F is an antideriv. of f.

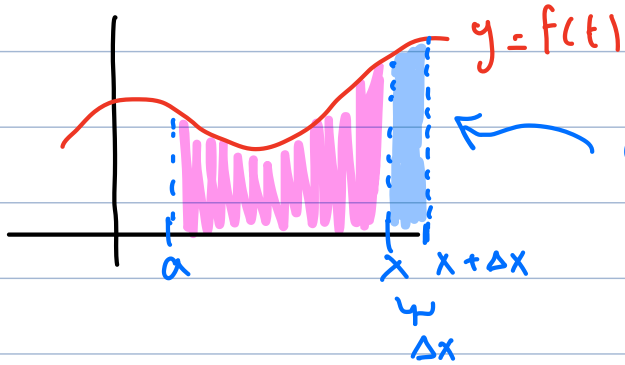
$F' = f$

FTC part 1

claim: in the setting above, $g'(x) = f(x)$

"proof": $g(x) = F(x) - F(a) \Rightarrow g'(x) = F'(x) - 0 = f(x)$

Alternative explanation in terms of area:



extra area $\approx$ $\underbrace{\text{length}}_{\Delta x} \cdot \underbrace{\text{height}}_{f(x)}$

$\overset{\shortparallel}{\Delta g}$

$$g'(x) \approx \frac{\Delta g}{\Delta x} \approx \frac{\Delta x \cdot f(x)}{\Delta x} = \underline{f(x)}$$

Summary of FTC: Let $f(x)$ be a <u>continuous</u> function

1) if $g(x) = \int_a^x f(t)\,dt$

   then $g'(x) = f(x)$

   (in other words, $\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$)

2) if $F'(x) = f(x)$

   then $\int_a^b f(x) = F(b) - F(a)$

Note: can think of FTC as saying derivatives and integrals are <u>inverse processes</u>

Note: continuous functions always have <u>antiderivatives</u>

Example Find $g(x) = \int_1^x t^2\,dt$

Answer: antiderivative: $\frac{1}{3}t^3$

$$\frac{1}{3}t^3 \Big|_1^x = \frac{1}{3}x^3 - \frac{1}{3}\cdot 1^3 = \boxed{\frac{1}{3}x^3 - \frac{1}{3}}$$

Example Find $\frac{d}{dx}\left(\int_2^x \sin(e^t)\,dt\right)$ (Hint: do not try to evaluate the integral)

Answer: use FTC I:

$$= \boxed{\sin(e^x)}$$

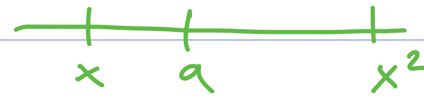
Example Find $\frac{d}{dx}\left(\int_x^2 \sin(e^t)\,dt\right)$

Answer:

$$= \frac{d}{dx}\left(-\int_2^x \sin(e^t)\,dt\right)$$

$$= -\underbrace{\frac{d}{dx}\left(\int_2^x \sin(e^t)\,dt\right)}_{e^x} = \boxed{-e^x}$$

Example $\dfrac{d}{dx}\left(\displaystyle\int_x^{x^2} \sin(e^t)\,dt\right)$



$$= \frac{d}{dx}\left(\int_x^a \sin(e^t)\,dt + \int_a^{x^2} \sin(e^t)\,dt\right)$$

$$= \underbrace{\frac{d}{dx}\left(\int_x^a \sin(e^t)\,dt\right)}_{\mathrm{I}} + \underbrace{\frac{d}{dx}\left(\int_a^{x^2} \sin(e^t)\,dt\right)}_{\mathrm{II}}$$

I: $\dfrac{d}{dx}\left(\displaystyle\int_x^a \sin(e^t)\,dt\right) = -\dfrac{d}{dx}\left(\displaystyle\int_a^x \sin(e^t)\,dt\right)$

$$= -\sin(e^x)$$

II: $\dfrac{d}{dx}\left(\displaystyle\int_a^{x^2} \sin(e^t)\,dt\right)$

$g(x) = \displaystyle\int_a^x \sin(e^t)\,dt$

$= \dfrac{d}{dx}\left(g(x^2)\right)$

FTC: $\underbrace{g'(x)} = \underline{\sin(e^x)}$

chain

$= g'(x^2)\cdot 2x = \sin\left(e^{x^2}\right) 2x$

Final answer: $\mathrm{I} + \mathrm{II} = \boxed{\sin\left(e^{x^2}\right)2x - \sin(e^x)}$

Example: Find $\frac{d}{dx}\left(\int_2^{\sqrt{\ln(x)}} e^{t^2}\,dt\right)$

$g(x) = \int_2^x e^{t^2}\,dt$ FTC: $\underline{g'(x) = e^{x^2}}$

$$= \frac{d}{dx}\left(g\left(\sqrt{\ln x}\right)\right) \overset{\text{chain}}{=} \underbrace{g'\left(\sqrt{\ln x}\right)}_{e^{\left(\sqrt{\ln x}\right)^2}} \cdot \underbrace{\frac{d}{dx}\left(\sqrt{\ln x}\right)}_{\frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x}}$$

$$e^{\left(\sqrt{\ln x}\right)^2} = e^{\ln x} = x$$

$$= \cancel{x} \cdot \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{\cancel{x}} = \boxed{\frac{1}{2\sqrt{\ln x}}}$$

Alternative perspective: use substitution

$$\frac{d}{dx}\left(\int_2^{\sqrt{\ln(x)} = u} e^{t^2}\,dt\right) = \frac{d}{dx}\left(\int_2^u e^{t^2}\,dt\right)$$

$$= \frac{d}{du}\left(\int_2^u e^{t^2}\,dt\right) \cdot \underbrace{\frac{du}{dx}}$$

FTC

$$e^{u^2} \cdot \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x}$$

$e^{(\sqrt{\ln x})^2}$

General rule: $\frac{d}{dx}\left(\int_{\ell(x)}^{u(x)} f(t)\,dt\right) = f(u(x))\cdot u'(x) - f(\ell(x))\cdot \ell'(x)$

<u>Indefinite integrals</u>. Let $f(x)$ be a continuous function. Then we will use $\int f(x)\,dx$ to denote <u>an antiderivative</u> of f

This is also called an <u>indefinite integral</u> of f

<u>Recall</u>: if F is an antiderivative of f, then so is $F+C$. So we leave the "$+C$" term in the antiderivative to remind ourselves the answer is not unique but can vary by a constant.

<u>Example</u>: $\int x^2\,dx = \frac{1}{3}x^3 + C$

<u>Note</u>: in contrast, a definite integral is an expression of the form $\int_a^b f(x)\,dx$, which is a number rather than a function.

Example: Find the indefinite integral

a) $\int (\underbrace{10x^4}_{10\frac{x^5}{5} = 2x^5} - \underbrace{2\sec^2 x}_{2\tan x})\,dx = \boxed{2x^5 - 2\tan x + C}$

b) $\int \frac{\cos\theta}{\sin^2\theta}\,d\theta = \int \underbrace{\frac{\cos\theta}{\sin\theta}}_{\cot\theta} \cdot \underbrace{\frac{1}{\sin\theta}}_{\csc\theta}\,d\theta = \int \cot\theta \csc\theta\, d\theta$

$= \boxed{-\csc\theta + C}$

Example: Find the definite integral:

a) $\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx$

$\frac{2x^4}{4} - \frac{6x^2}{2} + 3\tan^{-1}(x) \Big|_0^2 = \frac{x^4}{2} - 3x^2 + 3\tan^{-1}(x) \Big|_0^2$

$= \left(\frac{2^4}{2} - 3\cdot 2^2 + 3\tan^{-1}(2)\right) - \underbrace{\left(\frac{0^4}{2} - 3\cdot 0^2 + 3\underbrace{\tan^{-1}(0)}_{0}\right)}_{0}$

$= 8 - 12 + 3\tan^{-1}(2) = \boxed{3\tan^{-1}(2) - 4}$

b) $\int_1^4 \frac{2t^2 + t^2\sqrt{t} - 1}{t}\, dt$

First simplify

$$\frac{2t^2 + t^2\sqrt{t} - 1}{t} = \frac{2t^2 + t^{5/2} - 1}{t} = 2t + t^{3/2} - \frac{1}{t}$$

$$= \int_1^4 \left(2t + t^{3/2} - \frac{1}{t}\right) dt = t^2 + \frac{t^{5/2}}{5/2} - \ln t \Big|_{t=1}^{4}$$

$$= \left(4^2 + \frac{4^{5/2}}{5/2} - \ln 4\right) - \left(1^2 + \frac{1^{5/2}}{5/2} - \underbrace{\ln 1}_{0}\right)$$

$$= 16 + \underbrace{\frac{2}{5} \cdot \underbrace{\left(4^{1/2}\right)^5}_{32}} - \ln 4 - 1 - \frac{2}{5}$$

$$= \boxed{15 + \frac{62}{5} - \ln 4}$$

The following is a rephrasing of FTC 2:

Net change Theorem:

$$\int_a^b \underbrace{F'(t)}_{\text{rate of change}}\,dx = \underbrace{F(b)-F(a)}_{\text{net change}}$$

in words:
integral of rate of change
= net change

Applications: This applies to many physical quantities.

For example, if $F(t) =$ position, $= s(t)$
$F'(t) =$ velocity $= v(t)$

and so the net change Theorem states

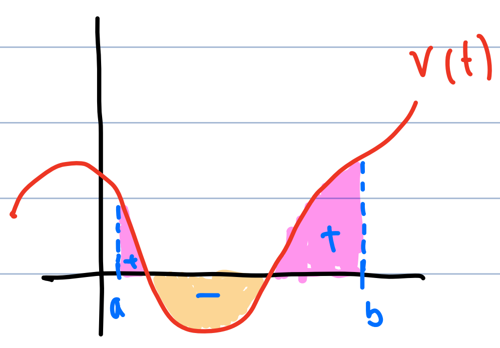
$$\int_a^b v(t) = s(b) - s(a) = \text{displacement from time } a \text{ to } b.$$

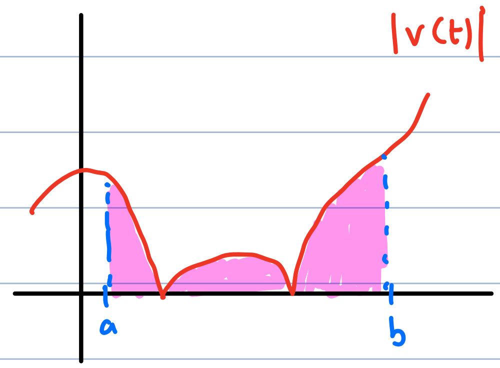
Note: if we want to find the total distance traveled rather than displacement, we can replace velocity with speed:

$$\text{speed} = |v(t)|$$

$$\text{total distance} = \int_a^b |v(t)|\,dt$$

Picture:





Example A particle moves along a line with velocity $v(t) = t^2 - t - 6$.

a) Find the displacement of the particle during $1 \le t \le 4$

b) Find the total distance traveled during that time period.

a) displacement $= \int_a^b v(t)\,dt = \int_1^4 (t^2 - t - 6)\,dt$

$$= \frac{t^3}{3} - \frac{t^2}{2} - 6t \Big|_1^4 = \left(\frac{4^3}{3} - \frac{4^2}{2} - 6\cdot 4\right) - \left(\frac{1}{3} - \frac{1}{2} - 6\right)$$

$$= \left(\frac{64}{3} - 8 - 24\right) - \left(\frac{1}{3} - \frac{1}{2} - 6\right)$$

$$= \frac{63}{3} - 38 - \frac{1}{2} = -17 - \frac{1}{2}$$

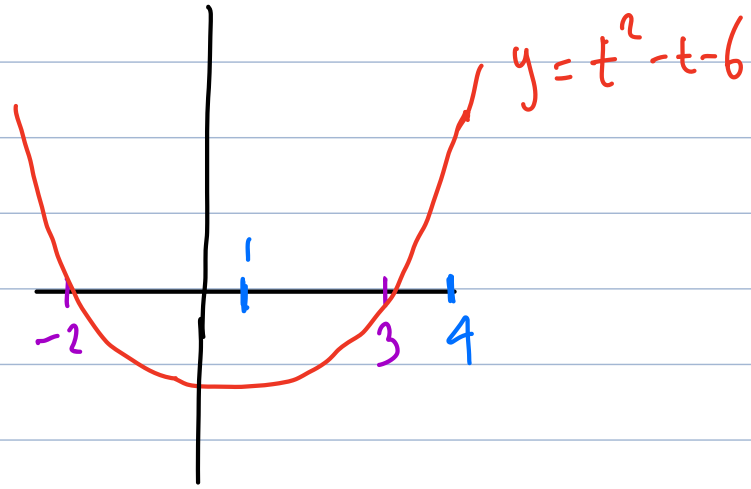
$$= -\frac{35}{2}$$

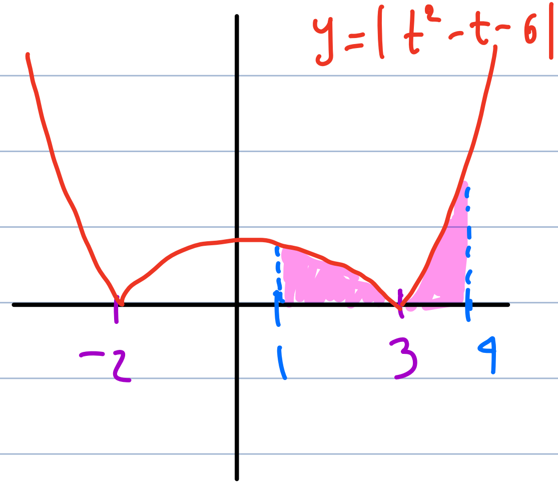
b) $v(t) = t^2 - t - 6$.

speed $= \int_a^b |v(t)|\,dt = \int_1^4 |t^2 - t - 6|\,dt$

$t^2 - t - 6 = 0$?

$(t-3)(t+2) = 0 \Rightarrow t = 3,\ t = -2$





$$|t^2-t-6| = \begin{cases} t^2-t-6 & \text{if} \quad t<-2 \\ -(t^2-t-6) & \quad -2<t<3 \\ t^2-t-6 & \text{if} \quad t>3 \end{cases}$$

$$\int_1^4 |t^2-t-6|\,dt = \int_1^3 |t^2-t-6|\,dt + \int_3^4 |t^2-t-6|\,dt$$

$$= \int_1^3 -(t^2-t-6)\,dt + \int_3^4 (t^2-t-6)\,dt$$

$= \ldots$ (take it from here)

Another perspective:

let $F(t)$ = antiderivative of t^2-t-6 = velocity

= position function

displacement = (position at $t=4$) − (position at $t=3$)

$$F(4) - F(1)$$

$$\text{total distance} = \underbrace{|F(3) - F(1)|}_{\text{moving left}} + \underbrace{|F(4) - F(3)|}_{\text{move right}}$$

$$F(1) - F(3) + F(4) - (F3)$$

other examples where "net change" can be applied:

- net change in altitude of a trail

  integral of slopes

- net change in population over time

  integral of rate of population growth

- total energy usage (megawatt-hours)

  integral of energy consumption

- total probability

  integral of probability density

- total mass of an object

  integral of density

- total revenue

  integral of marginal revenue

- net change in volume (say, in a lake or water tank)

  integral of rate of flow

Example Water flows from the bottom of a storage tank at a rate of $r(t) = 200 - 4t$ liters per minute, where $0 \le t \le 50$ (minutes)

Find the amount of water that flows from the tank during the first 10 minutes.

Answer: total $\underbrace{\text{flow}}_{\text{volume}}$ = $\int$ rate of flow

$$= \int_0^{10} \underbrace{(200 - 4t)}_{\text{liters/min}} \underbrace{dt}_{\text{min}}$$

$$= 200t - 4 \cdot \frac{t^2}{2} \Big|_0^{10}$$

$$= 2000 - 200 = \boxed{1800 \text{ liters}}$$